



CSJMU x IBM

BOB HACKS'26

Round 1 Submission

DATA DETECTIVES | The Green CSJMU Initiative

U.I.E.T, CSJMU, Kanpur

Team Details

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Problem Statement

CSJMU Sustainability Initiative Overview



Problem Statement

Cleanliness, waste management, water conservation, and electricity usage challenges with limited participation and no monitoring system.



Target Users

Students including hostel residents and day scholars, faculty members, and administrative and maintenance staff.



Beneficiaries

University gains efficiency, students and staff gain a greener campus, and the environment gains reduced consumption.

Current Challenges

Unnoticed waste

Garbage and waste remain unnoticed for long periods.

Resource wastage

Resource wastage (water, electricity) goes unreported.

Inefficient complaints

Manual complaint system is slow and inefficient.

Monitoring gaps

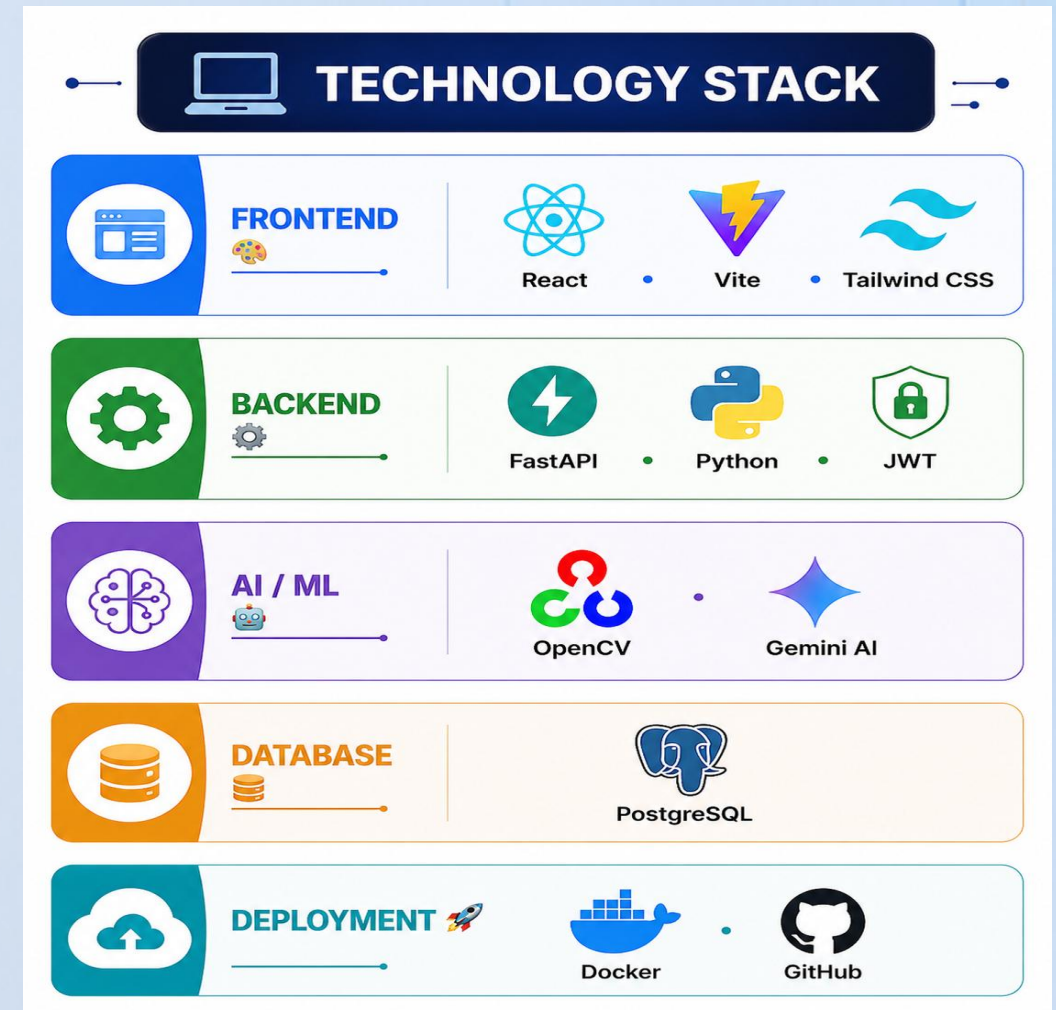
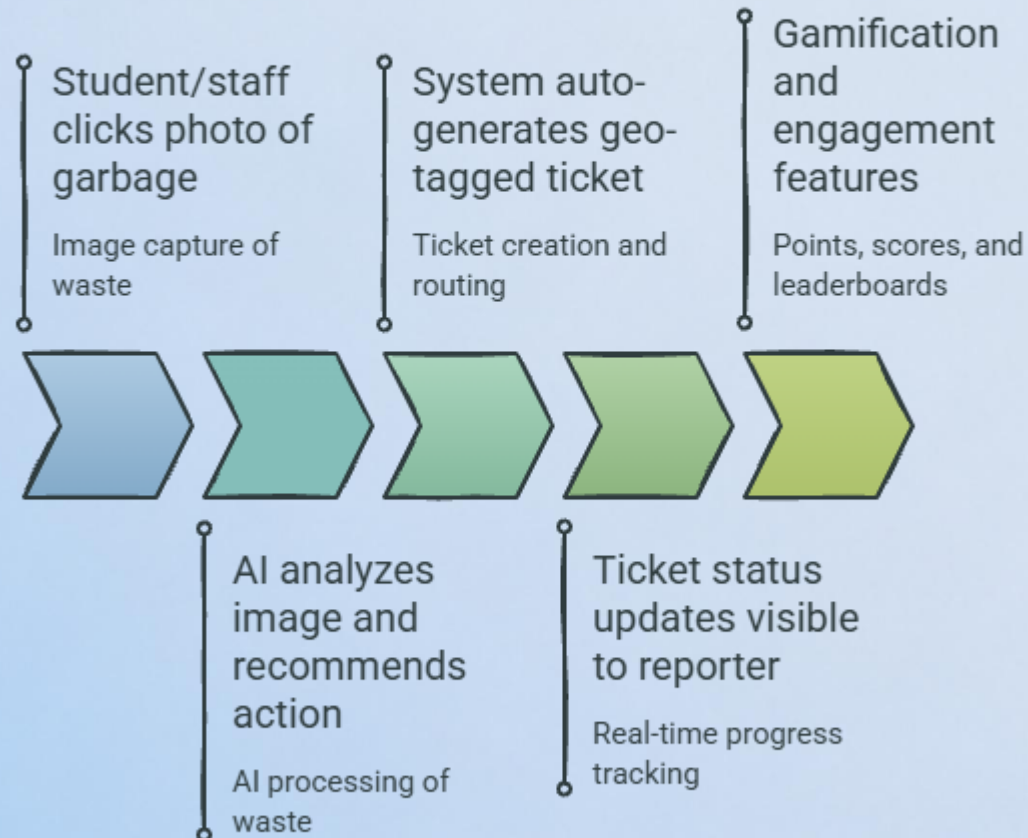
No centralized sustainability monitoring system.

Student engagement

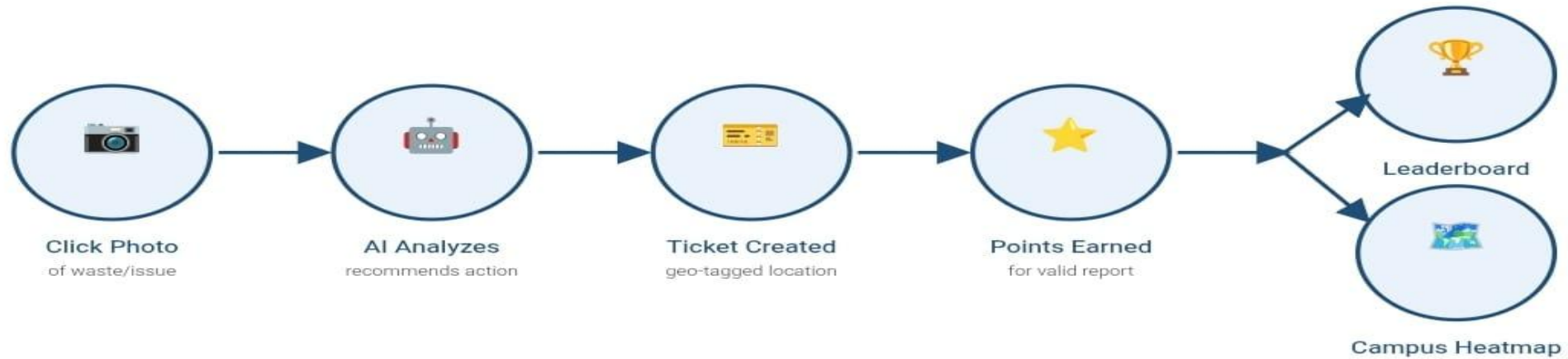
Lack of student engagement in keeping the campus clean.

Proposed Solution

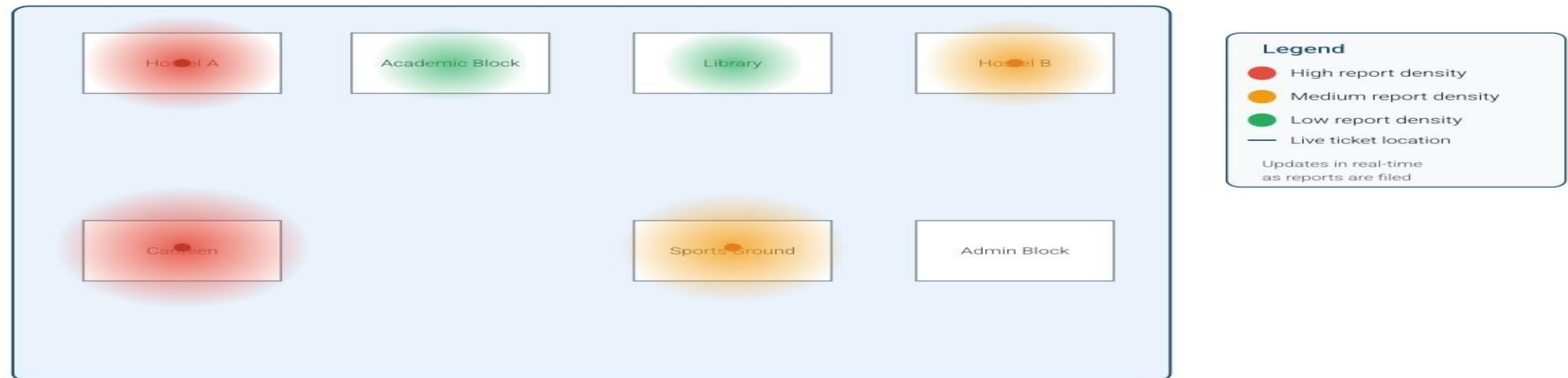
Campus Sustainability Reporting and Engagement Timeline



EcoCSJMU – Solution Workflow



Campus Waste Heatmap – EcoCSJMU



Concept visualization — actual heatmap generated dynamically from user-reported ticket locations

Wi-Fi occupancy monitoring

Anonymous Wi-Fi device counts identify low-occupancy spaces and reduce avoidable energy use.

LIVE · UPDATED 2 MIN AGO

Spaces monitored

Room 204 · Engineering Block

Expected: 45 people · AC & projector active

3

Wi-Fi devices

Energy-saving opportunity detected

Room 204 has had fewer than 5 connected devices for 18 minutes while appliances are marked active.

Send switch-off alert to operator

Seminar Hall A

Expected: 100 people · AV system active

76

Wi-Fi devices

Admin Office · Floor 1

Expected: 24 people · lights active

19

Wi-Fi devices

Hostel C Common Room

Expected: 35 people · fans active

28

Wi-Fi devices

POTENTIAL SAVINGS TODAY

8.6

kWh if flagged appliances are switched off

3 low-occupancy alerts raised

How it works

The system uses aggregate, anonymized Wi-Fi presence—never individual identity—to signal when a room may be empty. Operators verify the room before switching appliances off.

FEASIBILITY & VIABILITY



TECHNICAL FEASIBILITY

-  Built with proven technologies (React, FastAPI, PostgreSQL, TensorFlow, MobileNetV2)
-  AI model accurately classifies waste and detects issues
-  Geo-tagging, authentication and dashboards ensure scalability & security
-  Easily deployable on cloud or university servers



RESOURCE FEASIBILITY

-  Uses open-source technologies
-  Requires only smartphones and existing internet infrastructure
-  Integrates with existing departments without extra hardware
-  Low development and maintenance cost



OPERATIONAL FEASIBILITY

-  Simple and user-friendly for students, staff and administrators
-  Automated ticket routing reduces manual effort and response time
-  Gamification encourages participation and creates positive impact
-  Real-time analytics help prioritize cleanup and resource management



OVERALL

EcoCSJMU is technically feasible, cost-effective, scalable and operationally practical for smart campus sustainability.



Impact and Benefits



Who Benefits?



Students & Staff



University
Administration



Maintenance Departments



Geo tagging authentication
and dashboards ensure
scalability & security



Real-World Applications



Smart Campus
Management



Waste & Resource
Monitoring



Sustainable City
Initiatives



Social & Technical Impact



Cleaner and Greener
Campus



Faster Issue Resolution



AI-Driven Decision Making



Increased Community
Participation



Overall Impact | Promotes a smarter, sustainable, and environmentally responsible campus through AI-powered automation and citizen engagement.



Conclusion



EcoCSJMU shifts campus sustainability from passive awareness to active, community-driven action..



By combining a simple click-to-report mechanism with AI-driven insights, gamification, and real-time heatmaps. we turn every individual into a stakeholder in campus sustainability.



The solution directly closes CSJMU's core gap — the absence of monitoring and motivation — with a system that is low-cost, scalable, and easy to adopt:



Beyond compliance this creates a self-sustaining culture where sustainability becomes a habit, not a campaign.



Our vision: A campus that doesn't just look green. but behaves green — one photo, one point, one action at a time.



References & Links



Research References

- 1 IEEE Access (2021) — “Smart Campuses. Review & Challenges
- 2 Sciencebirect (2024) — “AI-Based Smart Waste Management”
- 3 ISPRS Annals (2025) — “Waste Management Using Artificial Intelligence”
- 4 IEEE Access (2022) — “IoT Enabled Smart Waste Marragement Systems”



Case Studies

- 5 Nanyang Technological University — AI Smart Campus.
- 6 RTS — AI in Waste Management Solutions



Tools & Technologies

- 7 Canva — Presentation Design
- 8 Nopkin A — Diagram Generation
- 9 IBM watsonx ai — AI Dotumentation & Reference



Additional Resources

- 10 IEEE Xolore
- 11 ScienceDirect
- 12 IBRS Dignal Liorary
- 13 IBBMwatsonx ai
- 14 IBM Watsonx ai Documentation